NIRANJANA GOPINATH

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OBJECTIVE

Final-year Computer Science student at Amrita Vishwa Vidyapeetham (CGPA 8.5) passionate about **AI/ML, intelligent systems, and IoT-enabled innovation**. Experienced in developing AI-powered applications and data-driven solutions, with a strong foundation in software engineering and backend development. Enjoy exploring the intersection of machine learning, connected devices, and scalable software to solve real-world challenges. Seeking roles where I can contribute to building practical AI solutions while growing as an engineer.

EDUCATION

B.Tech Computer Science, Amrita Vishwa Vidyapeetham, Coimbatore — Expected 2027

CGPA: 8.48

Relevant Coursework: Internet of Things, Data Structures & Algorithms, DBMS, OOP, Algorithm Analysis, Computer Organization, Operating Systems

SKILLS

Programming Languages	Python, C, C++, Java, MySQL, HTML, CSS
Libraries & Frameworks	**FastAPI**, **LiteLLM**, TensorFlow, Scikit-Learn, NumPy, Pandas, **spaCy**, **Hugging Fa**
Databases & Caching	**PostgreSQL**, **Qdrant** (Vector DB), **Redis**/Upstash, SQLite
Embedded & Hardware	**STM32**, **MAX30102**, **SX1278 LoRa**, I2C, **Edge ML**, C firmware, 433MHz RF
Tools & Platforms	**Langfuse**, **Docker**, GitHub Actions, Linux (Ubuntu), Zoho SalesIQ/Desk

WORK EXPERIENCE

AI Customer Support System — April 2026 – June 2026

Backend & ML Engineering Intern — *Remote*

- Architected an **AI customer support platform** integrating **Zoho SalesIQ**, **Zoho Desk**, and **OctoLens** into a unified **FastAPI** backend with message classification, auto-resolution, and ticket escalation.
- Engineered a **4-layer caching pipeline** (embedding, semantic, RAG, response) using **Redis** and **Qdrant**; achieved **100% cost savings** on repeated queries and **3–5s latency reduction** per cache hit.
- Built a **semantic classification cache** at **0.93 cosine similarity threshold**, catching equivalent queries (e.g. "how to login" vs "how do I log in") and eliminating redundant **LLM** calls.
- Implemented an async **LLM-as-Judge eval layer** across 5 dimensions; **95.1% classification accuracy**, **95.9% response relevance** across 115 evals; caught and fixed a **RAG hallucination** (faithfulness: **30% → 60%**).
- Designed **spike detection** with time-window aggregation (>20 issues/60 min → **Slack** alert) and cooldown deduplication to suppress notification spam during outages.
- Integrated **Langfuse** tracing with nested spans per message (classification, RAG, response, evals); failure-safe — Langfuse errors never affect the main pipeline.

PROJECTS

LoRa-Based Wearable Health Monitoring & Emergency Detection — *Team of 4*

Built a **LoRa-based IoT emergency monitoring system** using a **Tag → Relay → Gateway** architecture with **SX1278 (433 MHz)** communication over **2–10 km**. Developed **STM32 firmware** in C for real-time **BPM** and **SpO_2** monitoring using **MAX30102**, incorporating signal filtering and edge-based anomaly detection for sub-second emergency alerts. Improved reliability through **RSSI-driven relay selection**, automatic failover, and false-alarm suppression mechanisms.

Intelligent Document Processing & Analysis System: Architected an async document ingestion system using **FastAPI** and **PostgreSQL** with **REST APIs** for upload, **classification**, and **entity extraction**. Integrated **Hugging Face** and **spaCy** NER models achieving **89% entity extraction accuracy** on a 5,000-document corpus; implemented **pgvector**-backed semantic search delivering **2.4× lower query latency** vs keyword-only baseline.

Real-Time AIOps Monitoring System: Built a real-time monitoring platform using **FastAPI** and **Machine Learning** to ingest CPU/memory telemetry from **8+ hosts** at 1-second granularity. Deployed an **Isolation Forest** anomaly detection model achieving **91% precision** on fault injection; exposed **12 health KPIs** via an interactive **Chart.js** dashboard backed by **RESTful APIs**.

ACHIEVEMENTS

2nd Place — ICSRF 2025 SocioHack (National Social Hackathon) 2025
Team Arogya, Amrita Vishwa Vidyapeetham *Amritapuri, India*

- Selected among **Top 16 finalist teams** nationwide; won **1st place in Rural Healthcare** and **2nd place overall** across business, technology, and sustainability tracks.
- Led a cross-functional team to design, prototype, and present a scalable healthcare access solution aligned with social impact goals.